

Generation Interconnection System Impact Study Report

For Queue Project Q23-2354

Corners 230 kV
January 30, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2354. The Interconnection Customer proposes a 205.9 MW wind facility interconnecting at the Corners 230 kV substation in the Illinois area. The request was received on October 13, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 506 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,027,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bluebonnet-Russell Fork 230 KV line	Jones Chapel-Dekalb 500 KV	107.8	2,526	31.3
Ligonier-Cypress Crk 100 KV Line	Dekalb-Jones Chapel 500 KV L	106.5	432	5.2
Robison Park Substation-Russell Fork 230 KV	Dekalb-Jones Chapel 500 KV	105.5	3,415	42.6
Jones Chapel Substation-Dekalb Substation 500 KV	Robison Park-Russell Fork Substation 230 KV	104.9	5,380	65.3
Pueblo Xtra 500/230 kV Transformer Bank	Boca Raton-Pueblo Xtra 500 KV	104.2	2,553	41.9
Golden Gate-Ege 100 KV	Dekalb-Jones Chapel 500 KV	101.6	340	5.2
Cuba City-Ege 100 KV Line	Dekalb-Jones Chapel 500 KV L	101.4	355	5.2
Sta S Van Nuys-Gould 230 KV L	Troy Energy Llc-Richland Substation 500 KV	101.3	986	6.4
Farmersville-Kendallville 100 KV	Dekalb-Jones Chapel 500 KV	101.0	409	5.2
El Paso-Ganso 500 KV L	Lions Rd-Piney Creek Project 500 KV	100.7	3,648	13.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Russell Fork-Bluebonnet 230 KV line	Reconductor line	\$17,861,000	2% (\$272,000)
Cypress Creek-Ligonier 100 KV	Reconductor line	\$2,502,000	2% (\$41,000)
Robison Park Substation-Russell Fork 230 KV	Reconductor line	\$14,928,000	2% (\$230,000)
Deklb-Jones Chapel 500 KV	Reconductor line	\$42,819,000	2% (\$681,000)
Pueblo Xtra 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$35,708,000	2% (\$664,000)
Golden Gate-Ege 100 KV	Reconductor line	\$2,234,000	2% (\$37,000)
Cuba City-Ege 100 KV Line	Reconductor line	\$1,502,000	2% (\$25,000)
Sta S Van Nuys-Gould 230 KV L	Reconductor line	\$7,238,000	2% (\$117,000)
Kendallville Substation-Farmersville 100 KV	Reconductor line	\$1,979,000	2% (\$33,000)
El Paso-Ganso 500 KV L	Reconductor line	\$64,208,000	1% (\$927,000)
	Total allocated to project		\$3,027,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.